

Tumbleweed ROVER

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OBJECTIVE

We are building a scaled-down model of the Tumbleweed Rover that is set to explore Mars. This rover is an inexpensive way to further solar system exploration, as it uses wind energy to move and costs much less to produce and launch than traditional rovers.

$$2L + 2S = C$$

Total Connector Equation

$$L = kS$$

Long Connector Equation

$$S = \frac{C}{2k+2}$$

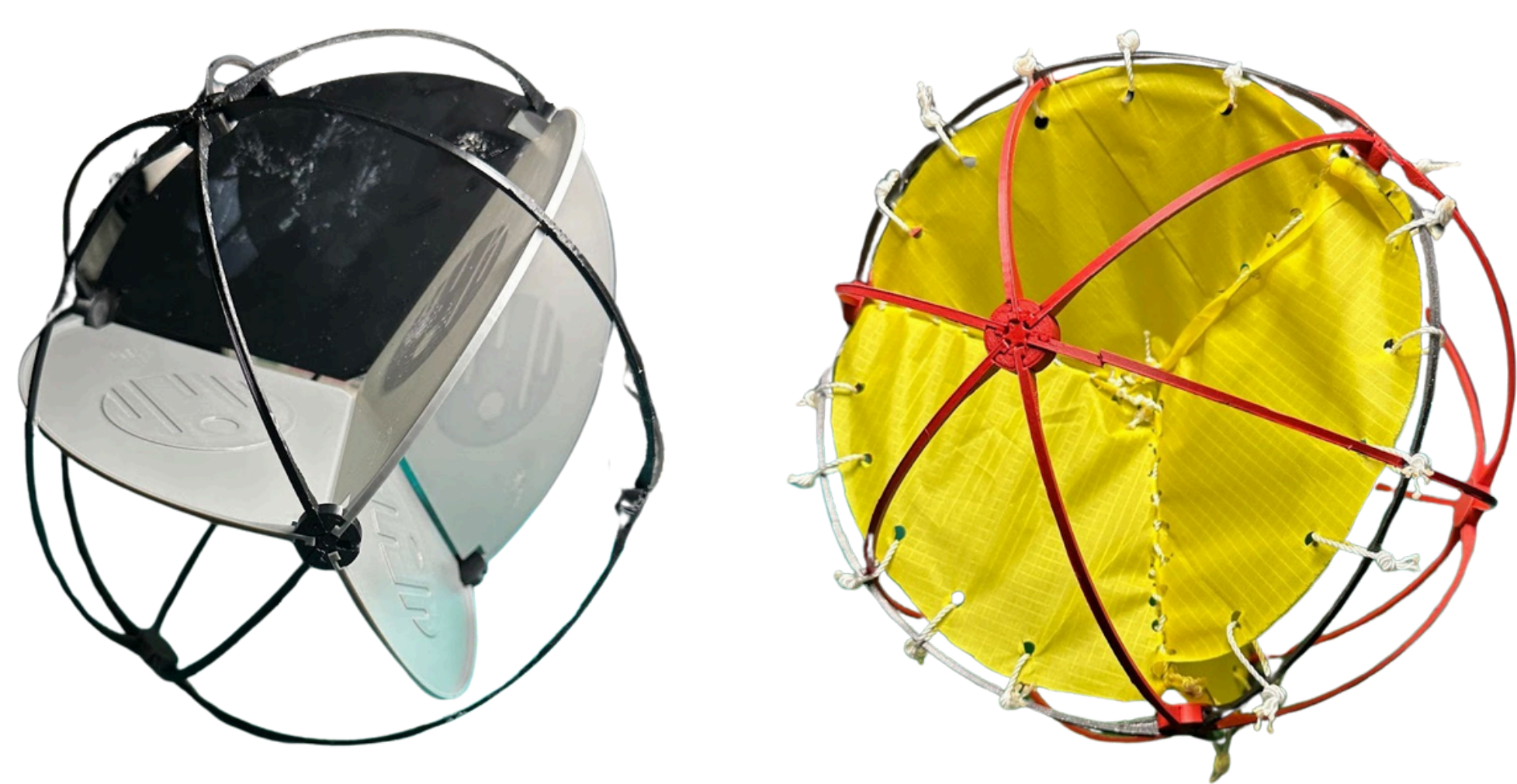
Short Connector Equation

$$D_E = D_M \left(\frac{g_M}{g_E} \right)^{\frac{1}{3}}$$

Gravity Scaling Equation

PLA
PROTOTYPE 1:
TEMPLATE

PLA
PROTOTYPE 2:
FABRIC SAILS



Blown by Wind

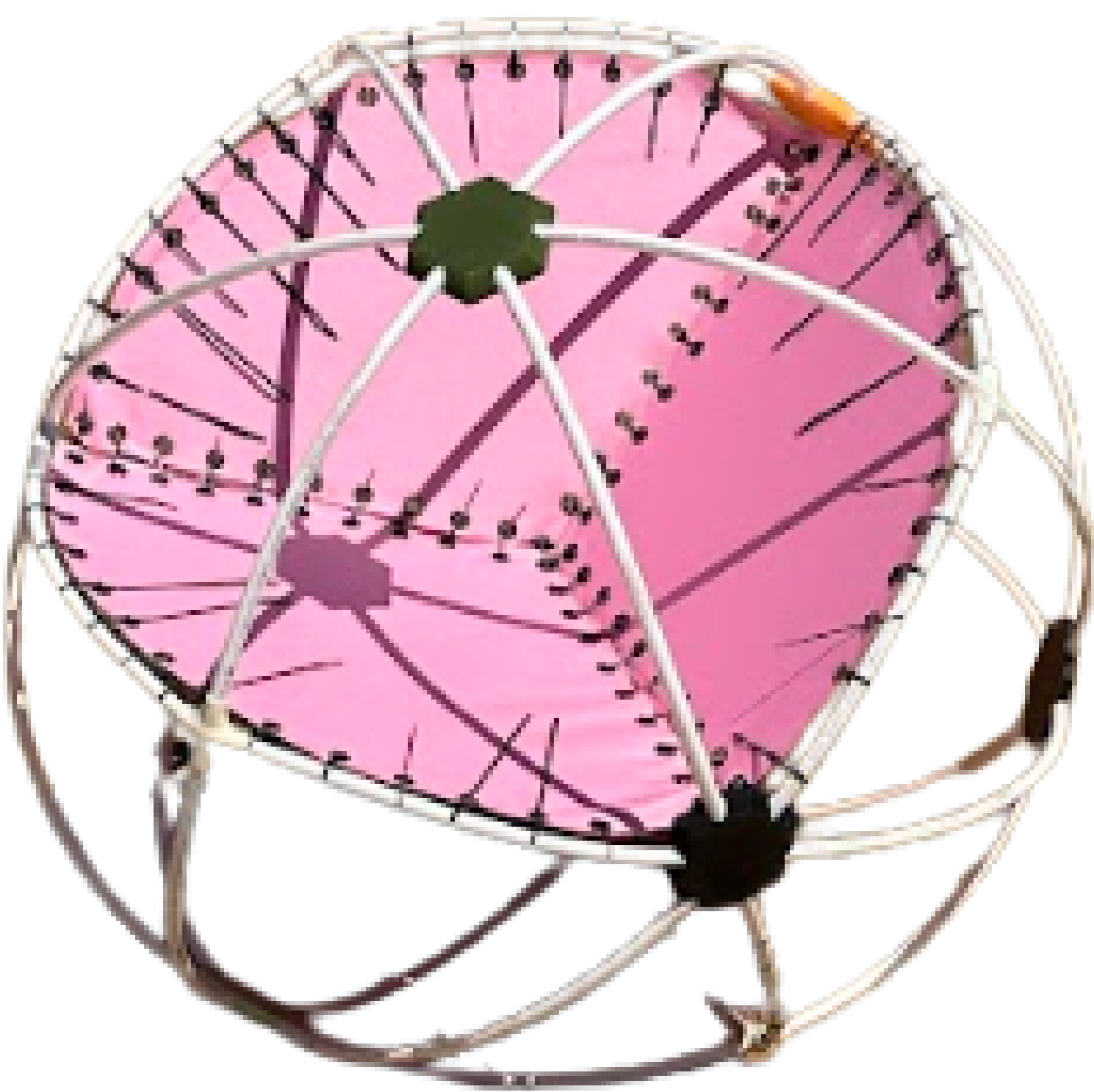
Travel Hundreds of Miles

More Data Collected

Modern and Easy

Cost-Effective

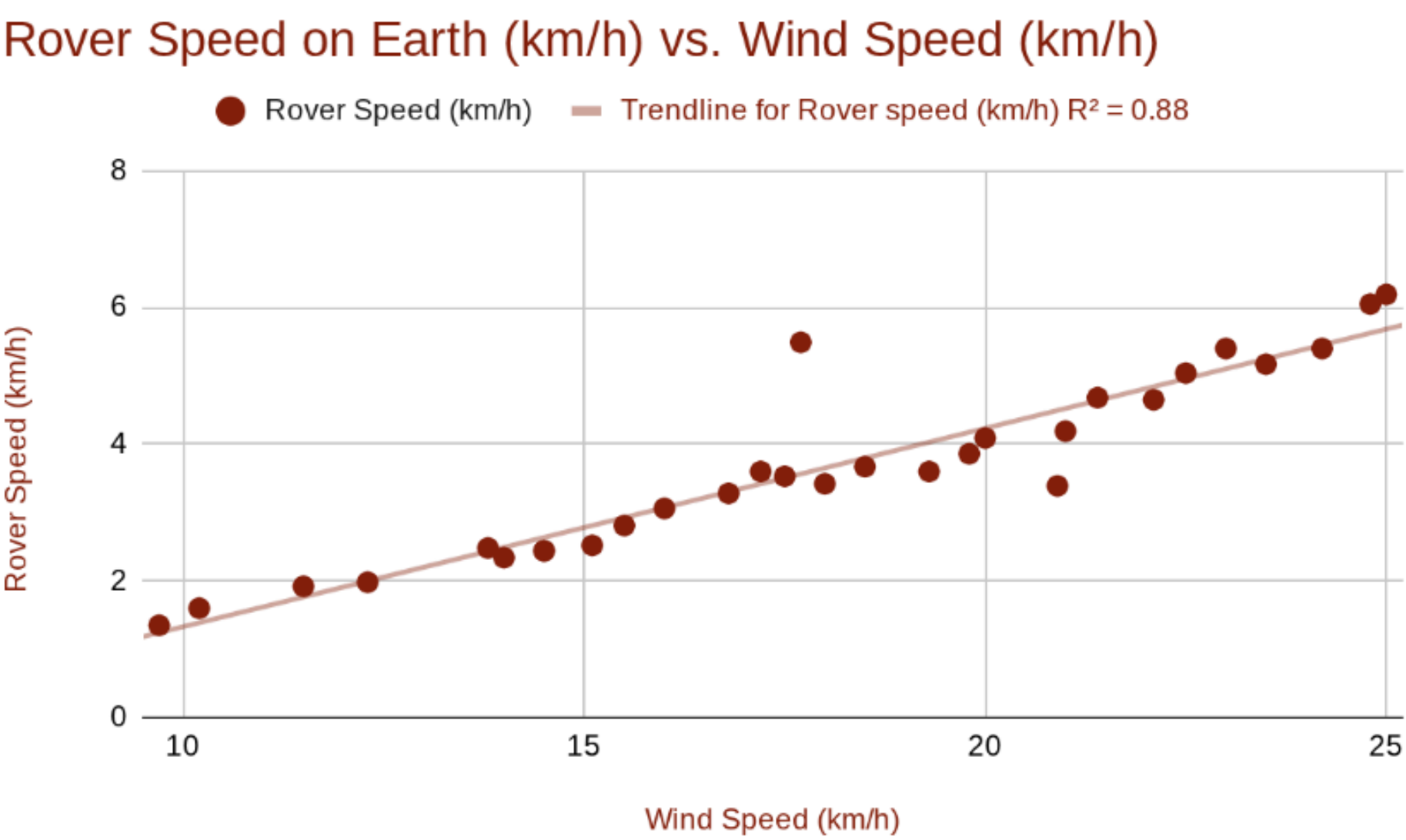
Lightweight



METHODS

1. Cut PEX-B tubing to length
2. 3D print 6-way and 4-way joints
3. Assemble frame
4. Cut six 50 cm radius fabric sails with central angle = 110°
5. Zip-tie sails together tetrahedrally
6. Attach sails to frame
7. Test rover recording distance traveled over time and wind speed
8. Convert data to Mars conditions

RESULTS



Mars Coefficient Calculation

$$F_{drag} = \rho \times v^2 \times K$$
$$K = \frac{M}{LT^2}$$
$$v_E \propto \sqrt{\frac{\mu_E g_E}{\rho_E}}$$
$$\frac{v_E^2}{v_M^2} = \frac{\mu_E m g_E}{K \rho_E} \times \frac{K \rho_M}{\mu_M m g_M}$$
$$\frac{v_E}{v_M} = \sqrt{\frac{\rho_M g_E \mu_E}{\rho_E g_M \mu_M}}$$
$$\frac{v_E}{v_M} = \sqrt{\frac{(0.020)(9.81)(1)}{(1.225)(3.721)(0.6)}} = 0.268$$

Earth Speed (km/h)	Mars Speed (km/h)
1.35	0.362
1.92	0.654
2.44	0.654
5.49	1.471